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Software Design & Analysis Document (SDAD)

Submitted to:

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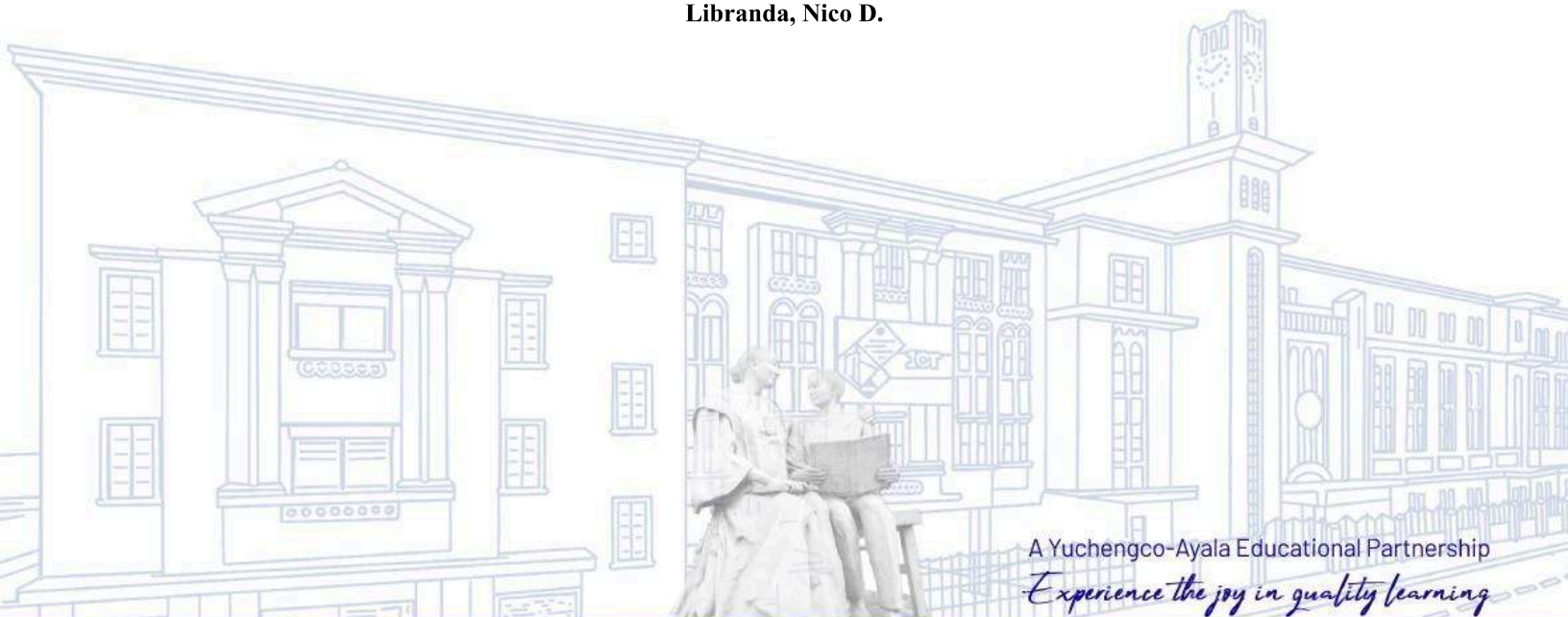
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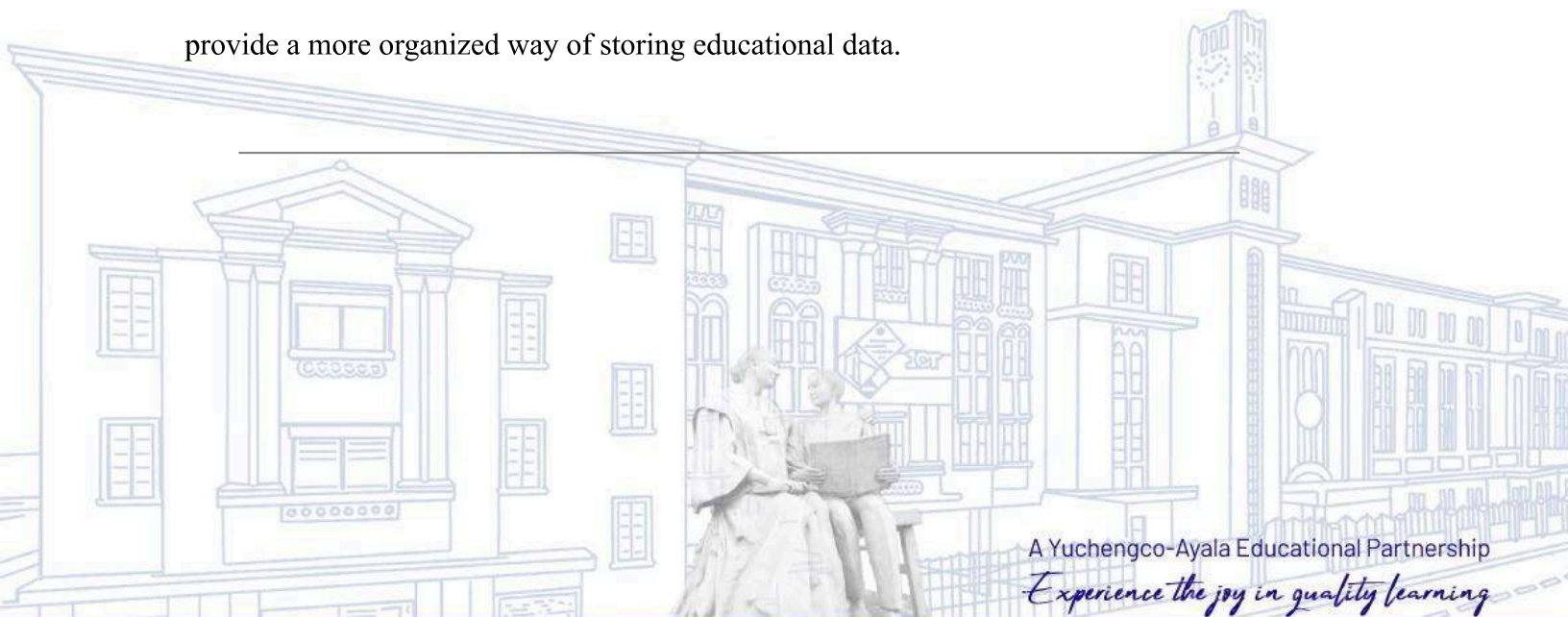
1. Introduction

1.1 Project Overview

The Student Resource Tracker System is a desktop application developed using VB.NET Windows Forms and Microsoft SQL Server. The system was designed to help schools and educational organizations efficiently manage learning resources such as books, modules, and educational materials.

The application allows users to perform CRUD operations (Create, Read, Update, and Delete) while using ADO.NET for database connectivity.

The main goal of the system is to improve resource management, reduce manual recording, and provide a more organized way of storing educational data.



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The Student Resource Tracker System is a VB.NET Windows Forms application connected to a Microsoft SQL Server database using ADO.NET. The main purpose of the system is to help schools, libraries, and student organizations manage educational resources efficiently.

The system allows users to:

- Add new learning resources
- View available resources
- Update resource information
- Delete unused or incorrect records
- Search for specific resources

The project uses CRUD operations (Create, Read, Update, Delete) connected to a SQL database called **StudentResourceTracker**.

This application was developed using:

- VB.NET Windows Forms
 - ADO.NET
 - Microsoft SQL Server
 - Visual Studio
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1.2 Chosen UN Sustainable Development Goal (SDG)

SDG 4 – Quality Education

The project supports SDG 4 by improving access to educational materials and helping schools organize learning resources more effectively.

Target:

“Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.”

The system helps educational institutions track books, modules, and learning materials in a simple and organized way.



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2. Requirements Analysis

2.1 Functional Requirements

Functional

Requirement	Description
1 Add Resource	The system shall allow users to add new resources into the database.
2 View Resources	The system shall display all stored resources using a DataGridView.
3 Update Resource	The system shall allow users to edit existing resource information.
4 Delete Resource	The system shall allow users to remove records from the database.
5 Search Resource	The system shall allow users to search for resources.



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|---|---------------------|--|
| 6 | Database Connection | The system shall connect to SQL Server using ADO.NET. |
| 7 | Auto Load Data | The system shall automatically load records when the form opens. |
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2.2 Non-Functional Requirements

No.	Non-Functional Requirement	Description
1	Usability	The interface should be simple and easy to use.
2	Reliability	The system should correctly save and retrieve data.
3	Performance	The system should load records quickly.
4	Maintainability	The code should be organized and readable for future updates.
5	Security	SQL parameters should be used to reduce SQL injection risks.



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|---|---------------|---|
| 6 | Compatibility | The system should run on Windows computers with .NET Framework installed. |
|---|---------------|---|
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3. Database Design

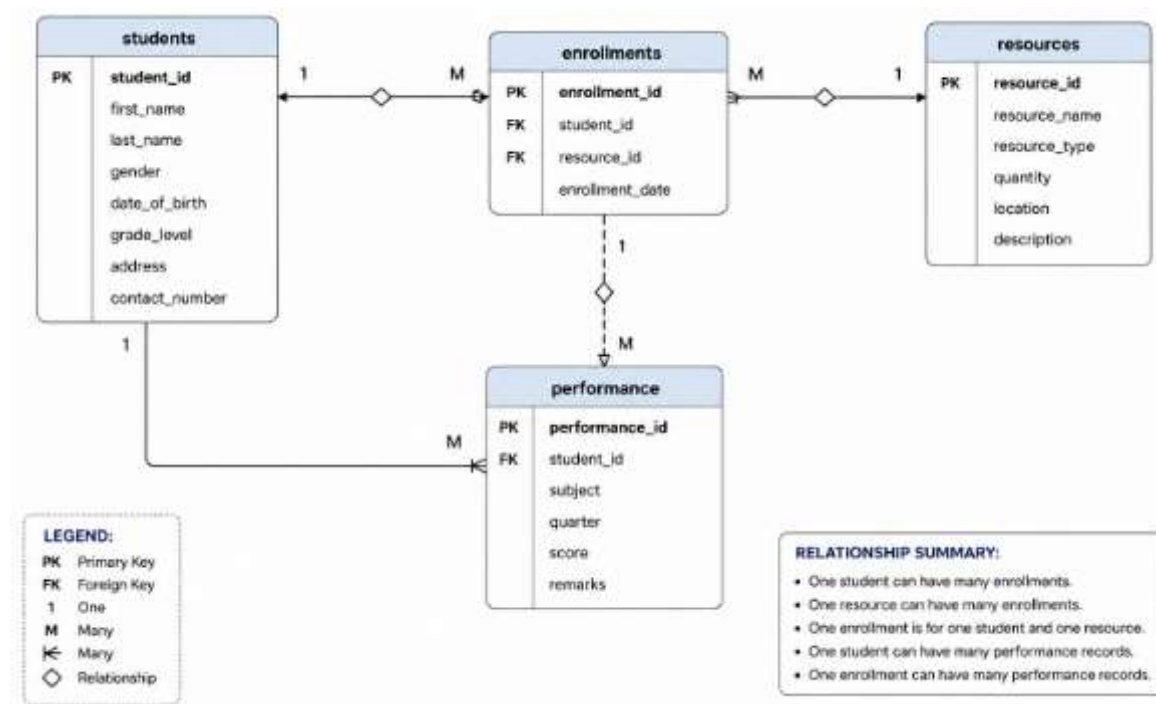
3.1 Database Name

StudentResourceTracker



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3.2 Entity Relationship Diagram (ER)





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3.3 Data Dictionary

Table: Resources

Field Name	Data Type	Constraints	Description
ResourceID	INT	Primary Key, Identity(1,1)	Unique ID of the resource
Title	VARCHAR(100)	NOT NULL	Name of the resource
Category	VARCHAR(50)	NULL	Type or category
Author	VARCHAR(100)	NULL	Author or creator
Quantity	INT	NULL	Number of available copies



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3.4 SQL Database Script

```
CREATE DATABASE StudentResourceTracker;
```

```
GO
```

```
USE StudentResourceTracker;
```

```
GO
```

```
CREATE TABLE Resources (ResourceID INT PRIMARY KEY IDENTITY(1,1),
```

```
Title VARCHAR(100),
```

```
Category VARCHAR(50),
```

```
Author VARCHAR(100),
```

```
Quantity INT
```

```
);
```

```
GO
```



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4. System Architecture

4.1 System Flow Overview

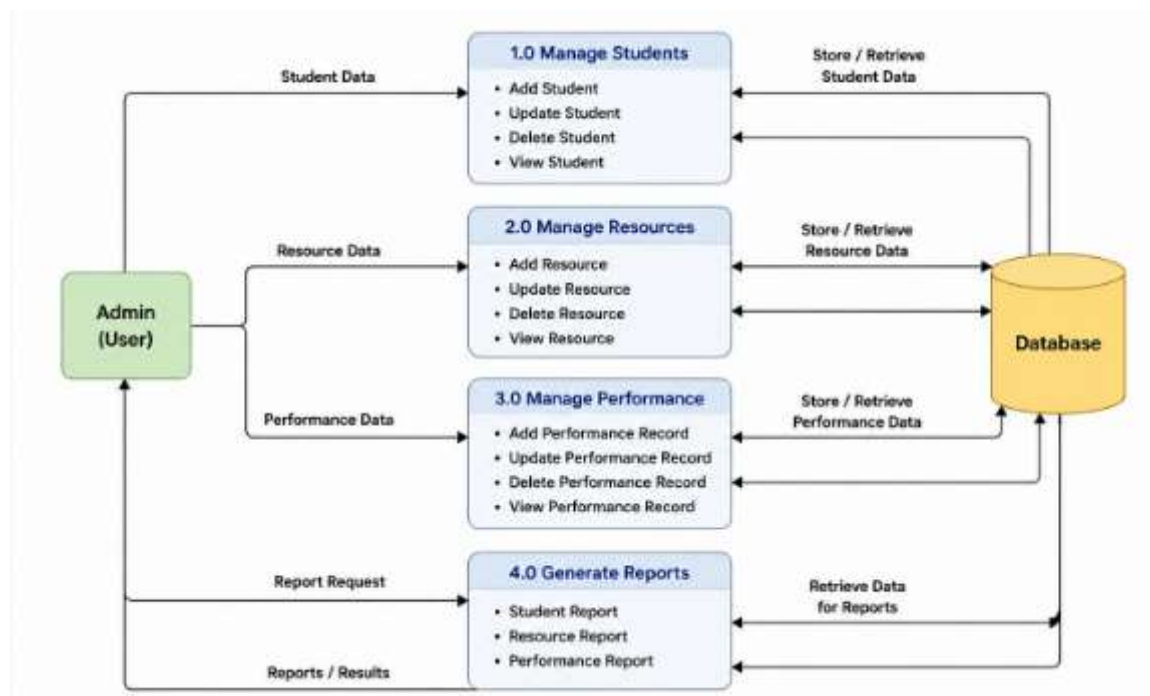
The application follows a simple client-server architecture:

1. User enters data in the VB.NET form.
 2. VB.NET application processes user actions.
 3. ADO.NET sends SQL commands to SQL Server.
 4. SQL Server stores and retrieves data.
 5. Results are displayed in the DataGridView.
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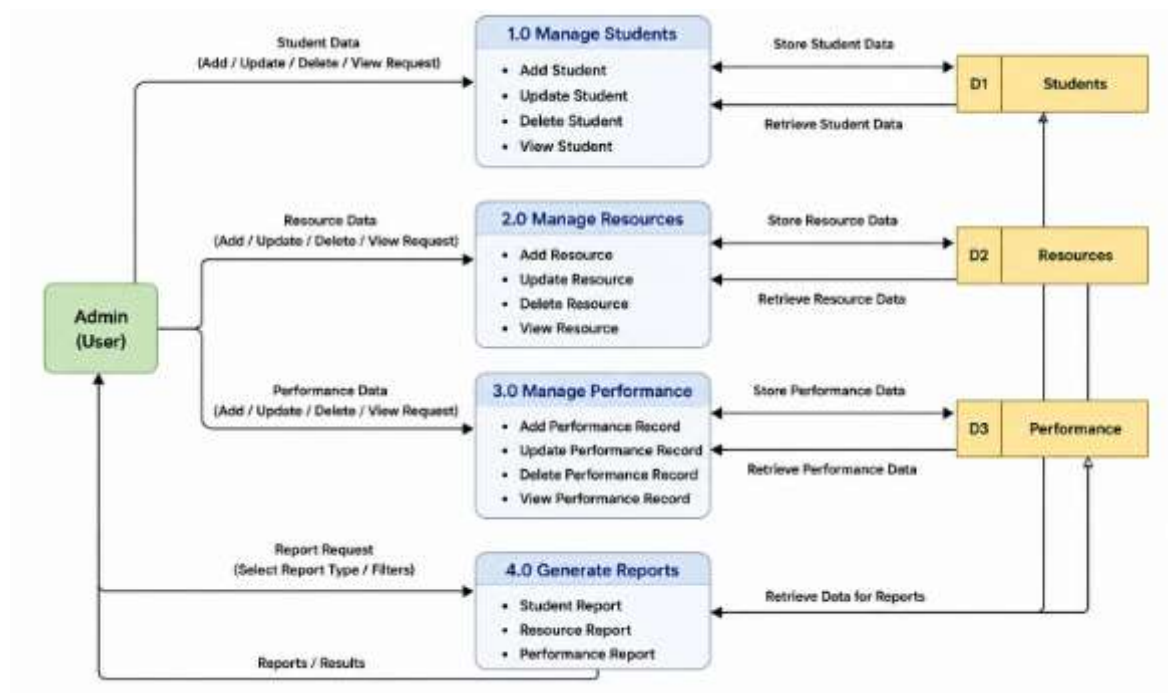
4.2 Data Flow Diagram (DFD)





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4.3 Data Flow Diagram (DFD Level 1)





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5. Source Code Analysis

5.1 Database Connection

The application uses a shared SQL connection:

```
Public Shared con As New SqlConnection(  
  
    "Data Source=(localdb)\MSSQLLocalDB;Initial  
    Catalog=StudentResourceTracker;Integrated Security=True"  
  
    )
```

Purpose:

- Connects the application to SQL Server.
 - Used by all CRUD operations.
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5.2 CRUD Operations

Add Resource

Uses INSERT SQL command to save new records.

Load Resource

Uses SELECT SQL command to display records in the DataGridView.

Update Resource

Uses UPDATE SQL command to modify existing data.

Delete Resource

Uses DELETE SQL command to remove records.

Search Resource

Uses SELECT query with conditions to find matching records.



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6. User Interface Design

6.1 Main Form Components

Component	Purpose
DataGridView	Displays all records
TextBox	Accepts user input
Buttons	Performs CRUD operations
Form	Main system window



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6.2 Buttons Used

Button	Function
Add	Saves new resource
Load	Displays all records
Update	Updates selected record
Delete	Removes selected record
Search	Searches for resources



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7. Individual Contributions

Group Member	Assigned Module/Form	Contribution
Cailan, Robert James V.	Database Design	Created SQL database and tables
Cailan, Robert James V.	VB.NET Form Design	Designed user interface
Blas, John Neo J.	CRUD Operations	Implemented Add, Update, Delete, and Load features
Magat, Miguel Ram	Documentation	Prepared SDAD and diagrams
Librandá Níco, D.	Testing	Performed debugging and testing

Note: Replace the placeholders above with your actual group member names.



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8. Testing and Evaluation

8.1 Test Cases

Test Case	Expected Result	Status
Add Resource	Data saved successfully	Passed
Load Data	Records displayed correctly	Passed
Update Resource	Selected data updated	Passed
Delete Resource	Record removed successfully	Passed
Search Resource	Matching results displayed	Passed



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9. Conclusion

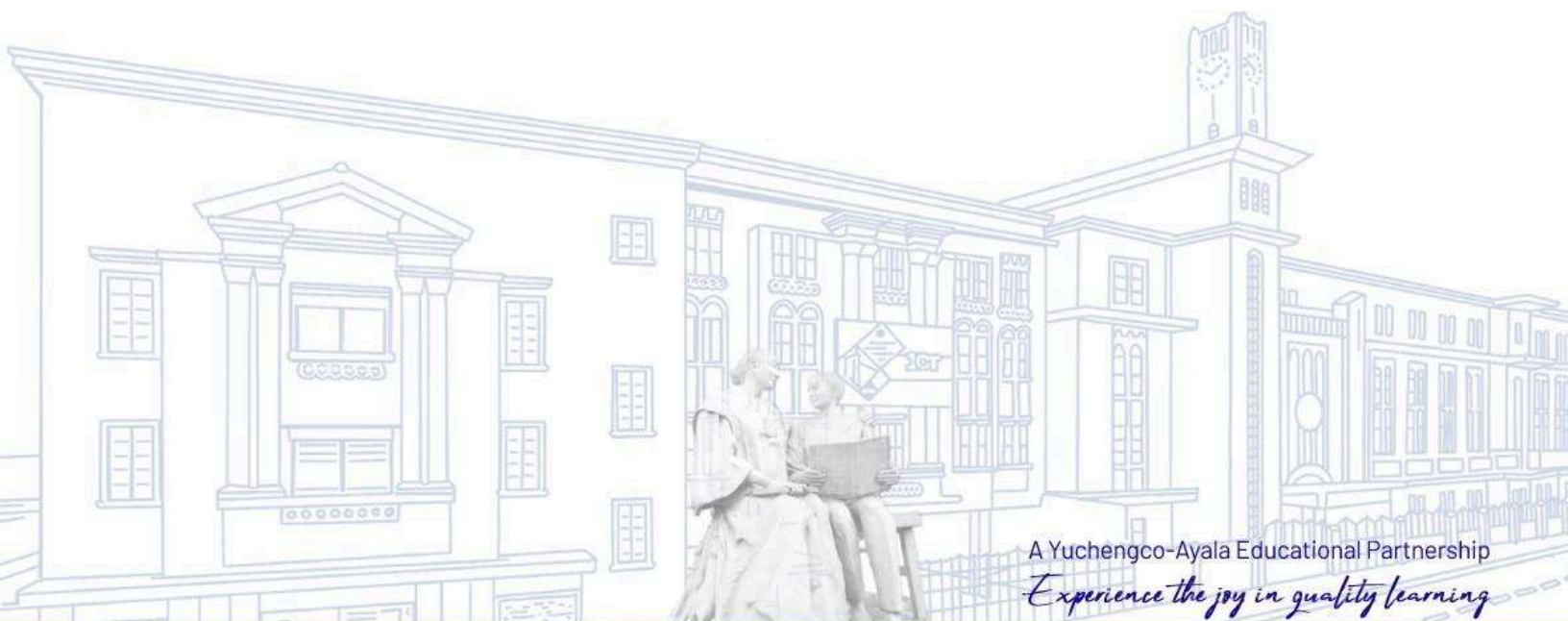
The Student Resource Tracker System successfully met the objectives of the project by creating a functional desktop application capable of managing educational resources efficiently. The system demonstrated the integration of VB.NET, ADO.NET, and SQL Server in implementing CRUD functionalities.

Additionally, the project supports SDG 4: Quality Education by helping educational institutions organize learning materials and improve accessibility to educational resources.

The project also enhanced the developers' knowledge and skills in database management, software design, and application development.



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